Budget: 100 kgCO_2 pro m Gurtlänge

Median

10%- / 90%-Fraktil

N_{Rd} [MN]

Zuggurt

Vollholz C24

Brettschichtholz GL24h

Baustahl S355

50 mm

100 mm

40 mm

14 mm

Stahlbeton C30/37, vorgespannt (10 MPa) ρ_p = 0.9%

u [mm]

100 mm

50 mm

32 mm

Betonstahl B500B (nackt)

Stahlbeton C30/37, ρ_s=0.6%

Umschnürter Beton C30/37 ρ_w=8.6%, ρ_l=0.6%

Druckgurt

Stahlbeton C30/37
Zug: ρ_s = 10%
A = 159'261 mm^2
g_{0k} = 4.7 kN/m
Druck: ρ_s = 0.6%
A = 430'563 mm^2
g_{0k} = 10.5 kN/m

Umschnürter Beton
ρ_w = 8.6%, ρ_l = 0.6%
A = 162'767 mm^2
g_{0k} = 5.1 kN/m

Brettschichtholz
GL24h
A = 1'000'000 mm^2
g_{0k} = 4.4 kN/m

Vollholz
C24
A = 1'818'182 mm^2
g_{0k} = 8.6 kN/m

vorgespannter Beton
σ_{cp} = 10 MPa
ρ_p = 0.9%
A_p = 3'239 mm^2
A_c = 358'461 mm^2
g_{0k} = 8.9 kN/m

Baustahl
A = 20'730 mm^2
g_{0k} = 1.6 kN/m

Betonstahl
A = 22'645 mm^2
g_{0k} = 1.7 kN/m